

STATE-OF-THE-ART REVIEW

Enzyme nomenclature and classification: the state of the art

 Andrew G. McDonald  and Keith F. Tipton

School of Biochemistry and Immunology, Trinity Biomedical Sciences Institute, Trinity College Dublin, Ireland

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Correspondence

 A. G. McDonald, School of Biochemistry and Immunology, Trinity College Dublin, Dublin 2, Ireland
 E-mail: amcdonld@tcd.ie

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The IUBMB enzyme classification system, available at the IUBMB ExplorEnz website, uses a four-component number (the EC number) that identifies an enzyme in terms of reaction catalysed. There were originally six recognized groups of enzymes: Oxidoreductases (EC 1), Transferases (EC 2), Hydrolases (EC 3), Lyases (EC 4), Isomerases (EC 5) and Ligases (EC 6). Of these, the lyases, which are defined as ‘enzymes that cleave C-C, C-O, C-N and other bonds by means other than by hydrolysis or oxidation’, present particular recognition and classification problems. Recently, a new class, the *Translocases* (EC 7), has been added, which incorporates enzymes that catalyse the movement of ions or molecules across membranes or their separation within membranes. A new subclass of the isomerases has also been included for those enzymes that alter the conformations of proteins and nucleic acids. Newly reported enzymes are being regularly added to the list after validation and where new information affects the classification of an existing entry, a new EC number is created, but the old one is not reused.

Introduction

Few biochemists are excited about nomenclature and terminology and even fewer would like to admit being so. In fact, it appears to be generally believed that if it is ignored it will go away. That is, of course, until it is found to be necessary to avoid ambiguity. Although we have all encountered people who believe the opposite, there are compelling advantages to knowing what you are writing or talking about and ensuring that others understand. It is that simple requirement that underlines the field of enzyme classification and nomenclature. It might be thought that all that was needed would be to ensure that enzymes had unambiguous identifiers, and the job was done. However, as new information becomes available and new enzymes are identified, the system requires continuous revisions and additions. For many years, textbooks and other

accounts stated that enzymes are divided into six classes, according to the type of reactions they catalyse, but recently it has proved necessary to add a new class, as well as some new subdivisions to the existing classes.

Why the enzyme list is needed

The need for a rational nomenclature for enzymes can be seen from the litany of unhelpful names for enzymes that abounded. Only the initiated few might have known what diaphorase might catalyse and how it might differ from DT diaphorase, or that the former was also called the old yellow enzyme, which of course was different from the new yellow enzyme (try EC 1.6.99.1 and EC 1.8.1.4). Similarly, the reaction

Abbreviations

EC (number), enzyme commission (number); IUBMB, International Union of Biochemistry and Molecular Biology; IUPAC, International Union of Pure and Applied Chemistry.

catalysed by rhodanese (thiosulfate sulfurtransferase; EC 2.8.1.1) was not apparent from its name and use of the name urokinase to describe a peptidase (EC 3.4.21.73) is confusing to the uninitiated, since we were all taught that kinases should add phosphate groups from ATP to their substrates. The chemists might guess that galactowaldenase catalysed a Walden inversion that changes the chirality of galactose. They could feel justifiably annoyed that galactose was not the substrate, but that enzyme (UDP-glucose 4-epimerase; EC 5.1.3.2) actually catalyses the interconversion of UDP- α -D-galactose and UDP- α -D-glucose. Confusion is further increased by the use of several different names for the same enzyme. Thus, for example, the quaintly named ali-esterase is apparently the same as monobutyrase, cocaine esterase, procaine esterase, methylbutyrase and B-esterase (carboxylesterase; EC 3.1.1.1). Similarly, ribonuclease T₁ (EC 4.6.1.24) has also been called barnase; bacterial ribonuclease Sa; guanyloribonuclease; *Aspergillus oryzae* ribonuclease; RNase N₁; RNase N₂; ribonuclease N₃; ribonuclease U₁; ribonuclease F₁; ribonuclease Ch; ribonuclease PP₁; ribonuclease SA; RNase F₁; ribonuclease C₂; binase; RNase Sa; guanyl-specific RNase; RNase G; RNase T₁; ribonuclease guaninenucleotido-2'-transferase (cyclizing) and ribonuclease N₁. Sometimes one finds that two different enzymes have ended up with the same name, for example, both D- and L-amino acid oxidases have been called ophio-amino acid oxidase, and they are also both yellow enzymes because they are flavoproteins. These and many other confusing enzyme names indicated the need for some way of unambiguously identifying enzymes; after all, it would be more than a little upsetting to have spent a considerable time working on cocaine esterase only to find that someone else had published the same results earlier with ali-esterase.

Many scientists still argue that precise nomenclature is unnecessary since they are well aware of the enzyme systems on which they are working. This thinking calls to mind the statement of Humpty Dumpty in Lewis Carroll's *Through the Looking Glass*, 'When I use a word it means just what I choose it to mean—neither more nor less'. Perhaps one example from a list of many will indicate that such an approach is not helpful. There were two families of amine oxidases, monoamine oxidase (EC 1.4.3.4) and EC 1.4.3.6 (now EC 1.4.3.21; primary-amine oxidase), which have different specificities, cofactor requirements and inhibitor sensitivities. Unfortunately, there have been several papers in which the authors have believed they were studying EC 1.4.3.4 but were in fact working with preparations of EC 1.4.3.6. Indeed, EC 1.4.3.6 was

marketed by one biochemical supplier as monoamine oxidase, for several years.

Although it is reasonable to want the name of an enzyme to give some indication of what it does, it has been suggested that the use of gene names might be used as an alternative [1]. This takes us back in time to the one gene per enzyme dictum of Beadle and Tatum [2], which is now known to be a gross oversimplification. In terms of enzyme activities, there are many cases of several gene products catalysing the same reaction, one gene product catalysing more than one distinct reaction as well as one reaction requiring the product of more than one gene. Thus, gene nomenclature and a function-based system, such as the IUBMB Enzyme List, should be regarded as complementary, serving different needs; one defining the thing and the other what the thing does. Furthermore, gene names are far from being unique, definitive, identifiers. For example, *selU*, *mmhH*, *ybbB* and *sufY*, are all recognized as being gene names for tRNA 2-selenouridine synthase (EC 2.9.1.3). Some databases, such as HUGO (<https://www.genenames.org/>), have proposed standard gene identifiers and names for proteins, including enzymes from humans and other vertebrates. The ExplorEnz enzyme list, however, includes data from many other species including plants, bacteria and archaea. The KEGG Enzyme database (<https://www.genome.jp/kegg/annotation/enzyme.html>), which is based on ExplorEnz, is a useful source of gene names that have been ascribed to the same enzyme from different sources, with further detail being available at the related KEGG Genes database. The NCBI Gene database (<https://www.ncbi.nlm.nih.gov/gene/>) contains an exhaustive list of gene identifiers, names and aliases, which can be searched by species.

Some order from chaos

In an attempt to produce a more rational system for enzyme classification, Otto Hoffmann-Ostenhof proposed that it could be based on the number of molecules involved in the reaction catalysed [3], whereas Malcolm Dixon and Edwin Webb argued that the defining feature of enzymes was their specificity and recommended a functional system that classified enzymes in terms of the reactions they catalysed. Their first attempt at a classification [4], then simply numbered from 1 to 659, divided the enzymes into three groups: (A) Hydrolysing Enzymes (enzymes 1–221), (B) Transferring Enzymes (enzymes 222–579) and (C) Other Enzymes reactions (enzymes 580–659). These were subdivided into broad categories, for example, the hydrolysing enzymes were subdivided into seven

subgroups depending on the type of link hydrolysed ranging from (A 1) hydrolysing -CO-NH- links, through hydrolysing amines (A 2) to hydrolysing other links (A 7). This system, although a valuable first attempt, had a number of obvious problems. The division into just three groups was crude and resulted in each one containing a category of enzymes ‘catalysing other reactions’. Even more of a problem was that of identifying a specific enzyme by a simple number, for example, if a new enzyme were added to the hydrolysing group, the number assigned to all enzymes in subsequent groups would have to be increased by 1.

For producing a more robust and useful system, Dixon and Webb, together with Hoffmann-Ostenhof, persuaded the International Union of Biochemistry [IUB; now renamed International Union of Biochemistry and Molecular Biology (IUBMB)] to establish an *Enzyme Commission* that developed the current system of classifying enzymes by serial numbers that were based on the overall reaction catalysed. The commission produced an interim report in 1961 [5], with its recommendations being also published in the second edition of Dixon and Webb’s book *Enzymes* [6]. The NC-IUBMB Enzyme list, or to give it its full title: *Recommendations of the Nomenclature Commission of the International Union of Biochemistry and Molecular Biology on the Nomenclature and Classification of Enzymes by the Reactions they Catalyse*, was updated in several printed editions, with the last appearing in 1992 [7] and also in the 3rd Edition of Dixon and Webb [8], with Edwin Webb having overall responsibility for its content. Further details of the history and development of this system have been published elsewhere [9,10].

After the publication of the 1992 volume, it was recognized that all subsequent material should be made available online. The book, which ran to over 850 pages, might be a useful reference source but was hardly riveting bedtime reading. Furthermore, that format limited the material that could be included in each enzyme entry and a delay of 5–10 years between publications was inadequate to deal with the pace at which newly recognized enzymes were being formally added to the list. Indeed, several new entries had been approved during the time that it took for the 1992 book to be published. Fig. 1 shows how the number of recognized and classified enzymes has increased since the first published list [4].

ExplorEnz: the enzyme database

Although several valuable databases had been produced that were based on the Enzyme list, including the ExPASy [11], KEGG [12] and BRENDA [13], members of the IUBMB enzyme committee had no direct responsibility for the accuracy of these or for ensuring that any new enzymes would be rapidly added to them. A flat-file version of the enzyme list was produced [14] in association with the IUBMB nomenclature page [15], but it had limited search capabilities and was not suited to curation by the enzyme nomenclature committee.

The ExplorEnz database (<https://www.enzyme-database.org/>) was developed as the official IUBMB enzyme classification and nomenclature website, with a separate curatorial interface [16,17]. Before being made available for public review at <https://www.enzyme-database.org/newenz.php>, new and revised enzyme

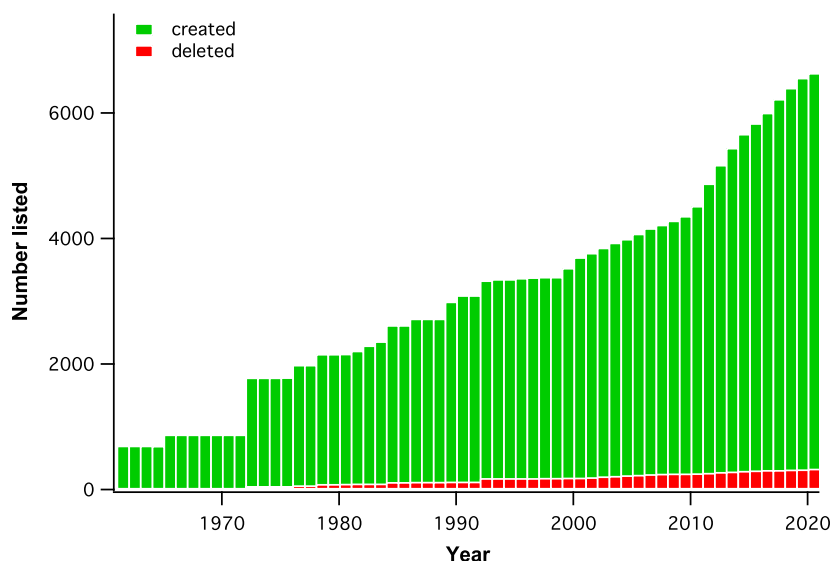


Fig. 1. Growth of classified entries in the Enzyme List. Shown as green columns are the numbers of new enzymes added to the IUBMB enzyme list per year, from 1961 to 2021. Deleted EC numbers, which are those entries whose removal has been necessitated by further evidence, are shown as red columns. Reclassified enzymes (transferred entries) are omitted.

entries that are proposed are evaluated by the IUBMB Enzyme Nomenclature committee, which currently comprises: Antje Chang, Julia Hoffmann and Ida Schomburg (Germany); Andrew McDonald and Keith Tipton (Ireland); Masaaki Kotera (Japan); Kristian Axelsen (Switzerland); Gerry Moss (UK); Ron Caspi (USA); with Ture Damhus (Denmark) providing essential guidance on IUPAC nomenclature issues. The internal review of proposed new entries is then followed by a 1-month public review period, after which entries are added to the official Enzyme List. The entire process of incorporating a proposed enzyme into the Enzyme List, including any further consultations with those who proposed the new entry, is a relatively slow process, typically ranging from 3 to 6 months in duration, but is deemed necessary to minimize the likelihood of the finally approved entries requiring further changes, unless necessitated by new data.

The Enzyme List is intended as a service to the community for enabling the identification of enzymes. It may be freely downloaded for use by others. At present, there are over 50 databases that use the EC data to a greater or lesser extent, although it is difficult to be precise since several new databases are announced with a published fanfare, only to become moribund and then vanish. Although IUBMB encourages other databases to acknowledge the source of the data they use, it cannot take responsibility for any errors in those databases.

Description of the enzyme list

What it contains

There are two criteria for inclusion in the enzyme list:

Criterion 1. An enzyme must be shown to catalyse a reaction that is different from any of those already listed. This classically entails identification of the substrates used and the products formed, which normally requires some purification of the enzyme so that it is free from any other reactions which might interfere with this process. This rigorous approach raises the obvious question of evidence from gene knock-out and knock-in experiments. A well-known example of why the results of such experiments alone should be treated with caution is that of lactose synthesis. *N*-acetyllactosamine synthase (EC 2.4.1.90) is converted into lactose synthase (EC 2.4.1.22) by binding α -lactalbumin [18]. Clearly, if anyone were able to do such an experiment, knocking out the gene for α -lactalbumin would remove lactose synthase activity and knocking it back in might be expected to result in its reappearance. There are several other examples of such activator proteins, for example,

β -*N*-acetylhexosaminidase (EC 3.2.1.52) requires the presence of a specific activator protein for hydrolysis of the ganglioside GM2 [19], and a number of multi-subunit enzymes requires noncatalytically active proteins for their assembly and activity regulation (e.g. [20–23]), necessitating some caution in the interpretation of the results of simple knock-out experiments.

Criterion 2. The enzyme must be physiologically occurring. Synthetic catalysts and enzymes that have been chemically or genetically modified to alter their activities or specificities are not included, although bacterial enzymes that are induced through growth on artificial substrates are. This offers the potential for using the Enzyme List for metabolic pathway construction, which would be undermined by inclusion of artificial enzymes, although the design and purpose of the system limit such an application [24,25]. It follows that the substrates should, whenever possible, also be compounds that an enzyme might encounter physiologically. Many studies on enzyme behaviour have used synthetic ‘unnatural’ substrates, simply because they facilitate assays, but it is helpful to indicate physiologically relevant substrates where possible. There remain several, otherwise well-characterized, enzymes where the physiologically important substrates have yet to be established. In many cases, such enzymes have such a broad specificity that it is appropriate to represent them by generic reactions, as discussed later.

Several systems postulate the existence of an enzyme from a gap in an experimentally determined metabolic pathway, sometimes ascribing an incomplete EC number to it [26,27]. Such interpretations should, however, be treated with caution (see, e.g. [28]), and the IUBMB takes a conservative approach that unless an enzyme has been shown definitively to exist, it will not be included because it will help nobody to include hypothetical enzymes. In addition, one, or more, papers describing the enzyme must have been published in the scientific literature, since it is important that users should be able to access such further information. If a request has been made for a new EC number and the pertinent data have been submitted for publication, or are ‘in press’, inclusion in the list is delayed until a citable paper has been published.

What it does not, yet, contain

Those of us with longer memories will recall being taught the axiom that all enzymes are proteins, but not all proteins are enzymes. Enzymes may need nonprotein parts to perform their catalytic functions, but the ‘backbone’ of the catalyst was always a protein. That was certainly the belief when the enzyme list was first produced.

Then came the *ribozymes* and the understanding that some enzymes were RNA, which may have protein attached but relied on a polynucleotide for its enzyme function [29–31]. They certainly do not have the catalytic versatility of the protein enzymes, being apparently restricted to modifying polynucleotide substrates, catalysing peptide bond formation and phosphate transfer, but nevertheless they can be highly specific biological catalysts and it would not seem appropriate to exclude them. The decision as to whether they should be included as a separate class or included within the current classification system is under discussion.

A long-standing problem concerns the *peptidases* (EC 3.4) [32], where basing classification and nomenclature on the reaction catalysed is inadequate. For example, if one were to take enzymes cleaving peptide bonds on the C-terminal side of an arginine, or lysine residue, that set would contain trypsin, papain, bromelain, some carboxypeptidases, cathepsins and caspases and many more enzymes. This problem was recognized when the first EC list was compiled, and it was decided that classifying these enzymes required a different approach. Some distinction can be made in terms of the catalytic mechanism involved, for example, the carboxypeptidases have been divided into the serine-type (EC 3.4.16), metallo- (EC 3.4.17) and cysteine-type (EC 3.4.18). Even that fails to address the complexity of these enzymes, for example, the endopeptidases that rely on an ‘active serine’ residue for catalysis (EC 3.4.21) currently contains 110 entries and this is far from complete. New peptidases are added to the list in response to requests, but the Enzyme List cannot be regarded as being in competition with the more comprehensive MEROPS database [33], which uses different classification criteria that are not subject to the same constraints as those of the IUBMB system.

The classification system

The information provided for each entry in ExplorEnz is concise, since the purpose is enzyme identification and not to provide a detailed account of its properties and functional involvements, which can be found elsewhere (e.g. [12,13]). Each entry contains the information summarized in Table 1 and elaborated, where necessary, in the following sections.

The EC number: classes and subclasses

The nomenclature system identifies enzymes according to a unique four-digit code, the Enzyme Commission, or EC, number. The first number (the class number) corresponds to the type of reaction catalysed. The six

Table 1. Contents of an Enzyme Entry. The fields provided are shown, with an accompanying description of their contents.

Field	Description
EC number	The four-component classification number that identifies the enzyme in terms of the reaction catalysed.
Accepted name	Usually, the most commonly used name for the enzyme, provided that it neither ambiguous nor misleading.
Reaction	The reaction catalysed presented, where possible, in a general form of the type: $A + B = P + Q$.
Diagram(s)	Where appropriate
Glossary	Where appropriate
Other name(s)	Any other names that have been used for the enzyme. This is as comprehensive as possible to aid searching for any specific enzyme. In cases where the same name has been given to more than one enzyme, this ambiguity will be indicated.
Systematic name	This describes the reaction catalysed in unambiguous terms.
Comments	Brief comments on the nature of the reaction catalysed, possible relationships to other enzymes, species differences, cofactor requirement, etc.
Links to other databases	To some other databases that may be helpful.
References	Some key references on the identification, nature, properties and function of the enzyme.
History	Creation date and any subsequent modification dates.

classes shown in Table 2 were recognized in the first Enzyme List and this division has persisted until 2018, when ‘Translocases’ were added as a new class (EC 7).

The type of reaction catalysed by most of these enzyme class should be self-explanatory and brief descriptions are available at <https://www.enzyme-database.org/class.php>. Ligases are defined as enzymes which catalyse the joining together of two molecules coupled to the breakdown of a diphosphate bond in a nucleoside triphosphate (NTP) such as ATP; these enzymes are often known as synthetases, or synthases. As discussed below, the lyases (EC 4) are perhaps more difficult to appreciate.

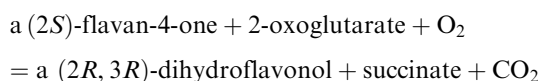
Each of these classes is further subdivided, with each enzyme being given a unique four-part EC number. The second part of the EC number refers to the subclass, which generally contains information about the type of compound or group involved. For the oxidoreductases, the subclass generally indicates the type of group in the donor that undergoes oxidation (e.g. EC 1.1 acts on the CH-OH group of donors, whereas EC 1.3 acts on the

Table 2. The classification of enzymes according to the type of reaction catalysed. The reaction catalysed column shows the general reaction scheme of the enzymes in each class. The new EC 7 Translocases class was added in 2018. II represents a membrane.

	Name	Reaction catalysed
1	Oxidoreductases	$^a\text{AH}_2 + \text{B} = \text{A} + \text{BH}_2$
2	Transferases	$\text{AX} + \text{B} = \text{BX} + \text{A}$
3	Hydrolases	$\text{A} - \text{B} + \text{H}_2\text{O} = \text{AH} + \text{BOH}$
4	Lyases	$\text{A} = \text{B} + \text{X} - \text{Y} = \text{A} - \text{B}$ $\begin{array}{cc} & \\ \text{X} & \text{Y} \end{array}$
5	Isomerases	$\text{A} = \text{B}$
6	Ligases	$\text{A} + \text{B} + ^b\text{NTP} = \text{A} - \text{B} + \text{NDP} + \text{P}$ (or $\text{NMP} + \text{PP}$)
7	Translocases	$\text{AX} + \text{B}_{[\text{side 1}]} \text{II} = \text{A} + \text{X} + \text{II}\text{B}_{[\text{side 2}]}$

^aWhere nicotinamide-adenine dinucleotides are the acceptors, NAD^+ and $\text{NADH} + \text{H}^+$ are used, by convention. ^bNTP = nucleoside triphosphate.

CH-CH group of donors). Enzymes that insert oxygen are grouped into EC 1.13 (the oxygenases) and EC 1.14, which includes many, but not all, enzymes commonly known as hydroxylases. Some enzymes in these groups, such as the flavoprotein monooxygenases [34], pose particular problems, since they are known to be active towards a number of ‘unnatural’ substrates, but the physiological substrates remain unknown. Where these enzymes catalyse generic reactions, they can be included as, for example, flavanone 3-dioxygenase (EC 1.14.11.9), which catalyses:



or long-chain alkane monooxygenase EC 1.14.14.28 (where long chain is defined in the ‘Comments’). There remain, however, a number of enzymes where there is insufficient information for their inclusion.

For the transferases, the second number indicates the type of transfer that takes place, with EC 2.1 indicating the transfer of one-carbon groups, EC 2.3 indicating the transfer of acyl groups, etc. The third number, which denotes the sub-subclass, further specifies the type of reaction involved. For the oxidoreductases, EC 1.-.1.- indicates that NAD or NADP is the acceptor, EC 1.-.2.- has cytochrome as the acceptor, etc. For the transferases, the sub-subclass gives more information about the group being transferred, with

EC 2.1.1 comprising the methyltransferases and EC 2.1.2 being the hydroxymethyl-, formyl- and related transferases. The fourth part is simply a serial number that is used to identify the individual enzyme within its sub-subclass.

This method of numbering, in addition to containing information about the nature of the reaction catalysed, makes it easy to accommodate a newly discovered enzyme by simply allotting the next available number at the end of its sub-subclass. Another principal is that any given EC number cannot be reused. Thus, if an enzyme were found to have been incorrectly classified, or not to exist, the EC number would be left unoccupied as a ‘deleted or transferred entry’, and a search for that EC number would indicate this and give a link to the new EC number, when appropriate.

Although computer programs have been developed for assigning EC numbers from the reactants involved [27,35], this should be, in the main, unnecessary since the design of the classification system should allow the first three components of the EC number to be deduced by simple inspection of the substrate(s) used and product(s) formed. The final component of the number must be the responsibility of the enzyme commission to avoid any confusion.

A consequence of using the reaction catalysed as the basis for classification is that all enzymes carrying out the same reaction are grouped together, even if they operate by different chemical mechanisms or occur as very different proteins in different species or cell types. Fructose-bisphosphate aldolase (EC 4.1.2.13), for example, includes enzymes working through different mechanisms. In the most recent update of the Enzyme List, this is dealt with by an explanatory comment, which reads: ‘The yeast and bacterial enzymes are zinc proteins. The enzymes increase electron-attraction by the carbonyl group, some (Class I) forming a protonated imine with it, others (Class II), mainly of microbial origin, polarizing it with a metal ion, e.g. zinc’. In the case of the alcohol dehydrogenases (EC 1.1.1.1), there are many isoenzymes in the human alone [4,36], and the enzyme from yeast is a very different protein. A similar situation applies to many other enzymes, such as the aldehyde dehydrogenases [34]. However, they are all classified together under one name and EC number. A significant difference in specificity is necessary for a separate EC number to be allocated.

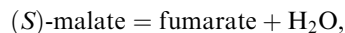
The lyases

These are defined as enzymes that cleave C-C, C-O, C-N and other bonds by means other than by hydrolysis

or oxidation. They usually differ from other enzymes in that two (or more) substrates are usually involved in one reaction direction, but there is one compound fewer in the other direction. When acting on the single substrate, a molecule is eliminated and this generates either a new double bond or a new ring. In accepted names, expressions like decarboxylase, aldolase, etc. are sometimes used. In cases where the reverse reaction is the more important, or the only one to be demonstrated, 'synthase' may be used in the name. Many lyases appear to catalyse an internal transferase reaction, forming a ring structure, accompanied by an elimination of part of the substrate, as illustrated in Fig. 2A.

When the atom in R^- that attacks X in the reactant is already adjacent to X , then the ring formed is two membered, that is, it is a double bond. In the reverse direction, the reaction involves breaking the ring, or double bond, by addition of the $^-Y-R'$ compound. Apparent complications can arise when the group

removed is water (a hydrogen ion plus a hydroxyl ion). Such enzymes are sometimes referred to as hydratases, hydro-lyases or, more often, simply called dehydratases. As is shown, for example, for the reaction catalysed by fumarate hydratase (EC 4.2.1.2),



such enzymes may be easily confused with hydrolases. The fact that they should be regarded as lyases depends on the details of the reaction catalysed and the fact that they are capable of catalysing the synthetic reaction [37], whereas simple hydrolase reactions would be expected to be essentially irreversible in aqueous solution (containing about 55.5 M water).

Ribonuclease was one specific enzyme that generated quite a lot of vituperative argument. It was originally classified as a transferase (EC 2.7.7.16), since it 'transfers the 3'-phosphate of a pyrimidine nucleotide

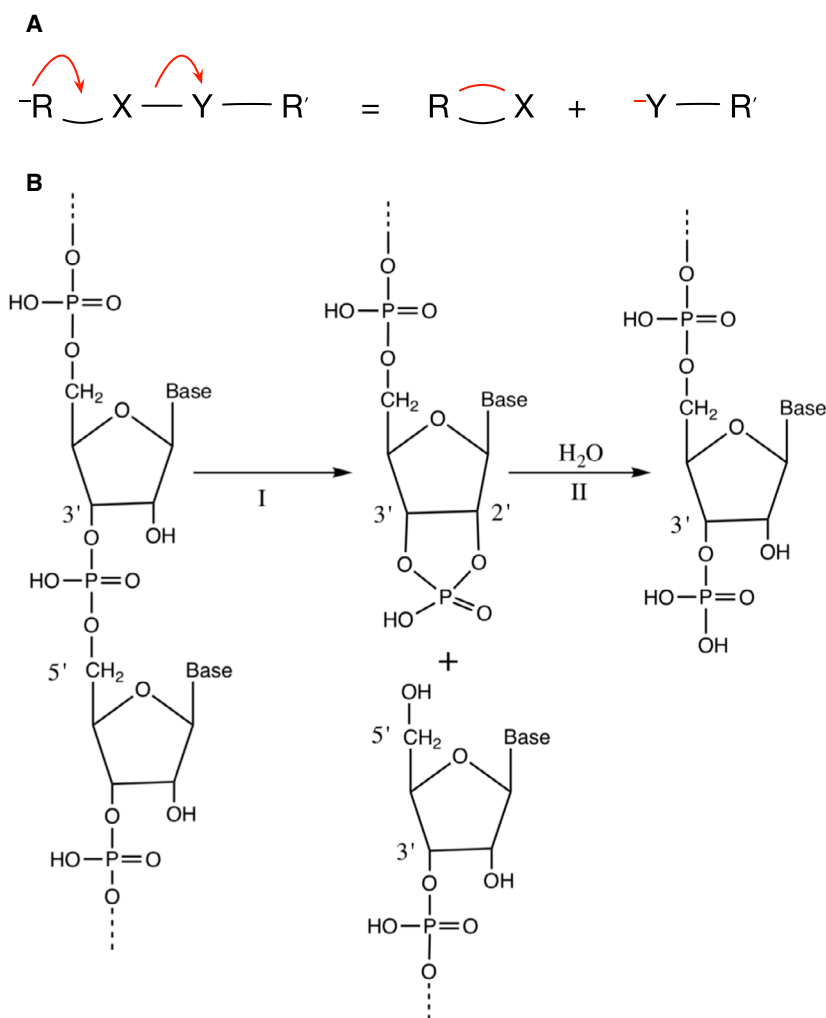


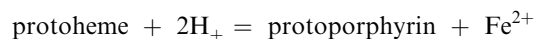
Fig. 2. Some Lyase mechanisms. A. General mechanism of internal transfer reactions. B. Mechanism of action of pancreatic ribonuclease (EC 4.6.1.18) in which an internal transferase reaction produces a 2', 3'-nucleotide phosphate, which is subsequently slowly hydrolysed to produce the 3' phosphate.

residue of a polynucleotide from the 5'-position of the adjoining nucleotide to the 2'-position of the pyrimidine nucleotide itself, forming a cyclic nucleotide', but then when it was realized that there were several RNases, it was renamed pancreatic ribonuclease. It then spent some time as a phosphoric diester hydrolase (EC 3.1.4.22) before settling amongst the hydrolases again, as one of the 'endoribonucleases producing other than 5'-phosphomonoesters' (EC 3.1.27.5), since it catalyses 'endonucleolytic cleavage to 3'-phosphomononucleotides and 3'-phosphooligonucleotides, ending in Cp or Up with 2',3'-cyclic phosphate intermediates'. Each of these classifications upset some ribonuclease aficionados, and pleased others, but they were all incorrect. The mechanism was established as long ago as 1966 [38]. As shown in Fig. 2B, the first reaction is an internal transfer that forms a cyclophosphate (reaction I). This product is actually released from the enzyme and the subsequent hydrolytic step (reaction II), which occurs by a reversal of reaction I with the hydroxyl group of water replacing the 5'-hydroxyl group of ribose, only occurring to any appreciable extent when the formation of the cyclophosphate product is essentially complete. Thus, the reaction is that of a lyase (a phosphorous-oxygen lyase) since it 'cleaves bonds by means other than by hydrolysis or oxidation'. This is the class of enzyme where two (or more) substrates are involved in one reaction direction, but there is one compound fewer in the other direction and when acting on the single substrate, a molecule is eliminated and this generates either a new double bond or, in this case, a new ring structure.

Because of this, pancreatic ribonuclease and some other ribonucleases have now been reclassified as phosphorous-oxygen lyases (EC 4.6.1.18). Those ribonucleases which appear to behave as simple hydrolases remain in EC 3 along with the DNases that cannot form 2',3'-cyclophosphate products. The previously used EC numbers now merely contain a note that the entry for pancreatic ribonuclease has been transferred to EC 4.6.1.18.

The lyases are divided into subclasses that depend on the type of bond being cleaved: EC 4.1 Carbon-carbon lyases, 4.2 Carbon-oxygen lyases, 4.3 Carbon-nitrogen lyases, 4.4 Carbon-halide lyases, 4.5 Carbon-halide lyases 4.6, Phosphorous-oxygen lyases and 4.7 Carbon-phosphorous lyases. That leaves a rather disparate group of enzymes catalysing reactions that did not fit into any of these subclasses, which were grouped into EC 4.99. Of the 12 enzymes currently in this subgroup, 8 are chelatas, which catalyse the insertion or removal of metal ions from

chelates, such as protoporphyrin ferrochelatase (EC 4.98.1.1), which catalyses:



It has been decided that this group of enzymes should be transferred to its own chelatase subclass of the lyases and this will occur once it has been publicly reviewed and approved.

Now we are seven

Earlier versions of the enzyme list contained a number of hydrolases that acted on nucleoside triphosphates, for example, in earlier versions of the enzyme list an enzyme 'Adenosine triphosphatase' was listed as EC 3.6.1.3. Subsequently, this single entry was expanded by indicating the function in the enzyme name, such as EC 3.6.1.35 (H⁺-transporting ATP synthase), EC 3.6.1.38 Ca²⁺-transporting ATPase, etc., while retaining their classification as hydrolases.

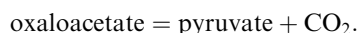
Each of these was shown as catalysing exactly the same reaction



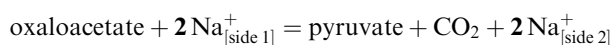
with what they actually did being amplified in the Comments field. This was far from being helpful and gave rise to several enzymes being listed as catalysing exactly the same reaction, although ATP hydrolysis was not their primary function. Purich [39] proposed that there should be a new class of enzymes, the *Energases*, which he defined as enzymes transducing covalent bond energy into mechanical work, mediating such processes as protein folding, self-assembly, G-protein interactions, DNA replication, chromatin remodelling and active transport. This helpful suggestion went a long way to resolving the problem catalysed by the simple hydrolase classification, but it would have resulted in the creation of a class of enzymes with quite disparate functions. As an interim measure, some enzymes were moved in 2000 to a new hydrolase subclass EC 3.6 'Acting on acid anhydrides to facilitate cellular and subcellular movement', with H⁺-transporting ATP synthase reclassified as EC 3.6.3.6. This, however, still kept these enzymes inappropriately listed amongst enzymes whose primary function was hydrolysis, so it was decided that a new class, EC 7 *Translocases*, was needed. These are defined as 'enzymes that catalyse the movement of ions or molecules across membranes or their separation within membranes'. The reaction is designated as a transfer from 'side 1' to 'side 2' because the designations 'in' and 'out' (or 'cis' and 'trans'), which had

sometimes been used previously, lack clarity and can be ambiguous. The Comments associated with each entry then describe the specific translocations catalysed.

A number of other enzymes whose function is transport were also recognized, for example, NADH:ubiquinone reductase (Na^+ transporting) had been classified as EC 1.6.5.8. Oxaloacetate decarboxylase was originally classified as a hydrolase EC 4.1.1.3, catalysing the reaction



It was subsequently recognized that there were two distinct enzymes, oxaloacetate decarboxylase (EC 4.1.1.112), which simply catalyses the reaction shown above and oxaloacetate decarboxylase (Na^+ extruding), now EC 7.2.4.2, which catalyses



The subclasses in the translocases classification designate the types of components transferred and the sub-subclasses indicate the reaction processes that provide the driving force for the translocation, as summarized in Table 3.

Exchange transporters that are not dependent on enzyme-catalysed reactions, such as the exchange of ions across membranes, are not included, while pores that change conformation between open and closed

Table 3. Subdivisions within the translocase class (EC 7). The new translocase class (EC 7) is divided into the subclasses shown based on the type of compound being moved, with the sub-subclass indicating the reaction that provides the driving force for the translocation.

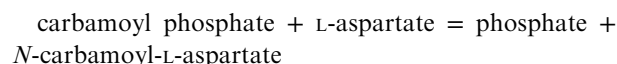
Subclass (EC)	Sub-subclass (catalysing)	Sub-subclass – the driving force
7.1	Translocation of hydrons ^a	7.x.1 – linked to oxidoreductase reactions
7.2	Translocation of inorganic cations and their chelates	7.x.2 – linked to nucleoside triphosphate hydrolysis
7.3	Translocation of inorganic anions	7.x.3 – linked to hydrolysis of a diphosphate
7.4	Translocation of amino acids and peptides	7.x.4 – linked to a decarboxylation reaction
7.5	Translocation of carbohydrates and their derivatives	
7.6	Catalysing the translocation of other compounds	

^aHydron is the general name for H^+ in its natural abundance.

states in response to phosphorylation or some other catalysed reaction are now classified under EC 5.6 (Macromolecular conformational isomerases).

The reaction catalysed

This is usually described in words as a balanced equation, with no attempt to balance reactions in terms of charge. For example, the reaction catalysed by aspartate carbamoyltransferase (EC 2.1.3.2) is written as:

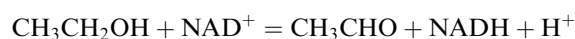


When it is not practicable to describe the reaction as a simple equation, a brief description is used, for example, the reaction for fructan β -fructosidase (EC 3.2.1.80) is given as:

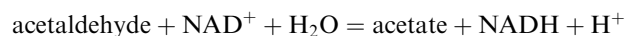
Hydrolysis of terminal, nonreducing (2 $\rightarrow$ 1)- and (2 $\rightarrow$ 6)-linked β -D-fructofuranose residues in fructans.

Two specific features of the reactions shown should be emphasized:

- 1 *No indication of the reaction direction.* It should be noted that the direction in which the reaction believed to operate *in vivo*, or, indeed, whether it is freely reversible is not specified (i.e. an equals sign is used rather than arrows). By convention, the direction chosen for the reaction is the same for all enzymes in any given sub-subclass. Although this might seem to be less helpful than it might be, it must be remembered that, for some enzymes, such as lactate dehydrogenase, glutamate dehydrogenase [NAD(P)^+] (EC 1.4.1.2) and fructose-bisphosphate aldolase (EC 4.1.2.13), the preferred reaction direction varies with cellular conditions. Furthermore, the equilibrium constant of the reaction may be misleading in terms of the direction in which it actually operates *in vivo*. As a well-known example, the equilibrium oxidation of ethanol by alcohol dehydrogenase (EC 1.1.1.1):



very much favours ethanol formation under physiological conditions, but ethanol oxidation is the dominant direction *in vivo* because acetaldehyde (ethanal) is rapidly removed in the essentially irreversible reaction catalysed by aldehyde dehydrogenase (NAD^+ ; EC 1.2.1.3)

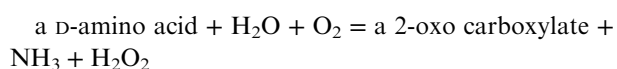


Thermodynamic data for many enzymes can be found in the *NIST Thermodynamics of Enzyme-*

catalysed Reactions database [40] and kinetic data are included in the BRENDA database [13], which may provide detail to determine reaction equilibria through Haldane relationships. It should be emphasized, however, that only data that refer to 'physiologically relevant conditions' should be considered and that it is the thermodynamic properties of the overall metabolic system, not of the individual reaction, that are important in determining the flux direction (see [9,10]).

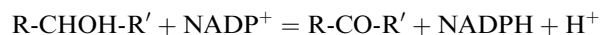
2 Charge balance is not considered. Generally, the charge on larger molecules is not indicated with that on inorganic anions and cations being used for convenience. Although some presentations have aimed to show charge-balanced reactions, assuming that all enzymes operate at the same pH, often assumed to be 7.2, this can be grossly misleading, since it takes no account of source or metabolic conditions. Although the cytosolic pH in many mammalian cells is generally taken to be about 7.2–7.4 under 'normal' conditions, that in the Golgi is about 6.4 [40–42]. In cardiac mitochondrial matrix, under normal, beating conditions, the pH is 7.91, whereas that in HeLa cells it is 7.98 [43]. In bacteria (both Gram-negative and -positive), the intracellular pH is affected by the pH of the media in which they were maintained [44]. Furthermore, intracellular pH is dynamic, for example, both the cytosolic and mitochondrial pH values will fall in hypoxic conditions [41,43] and the intracellular pH in yeast has been reported to fall from about 6.5 to 5.5 following a glucose or ethanol pulse to glucose-limited chemostat cultures [45]. Free phosphate presents an insuperable problem since its charge is likely to be fractional. One of the three pK_a values of free phosphate is 6.82, so its actual charge state will vary with the metabolic conditions and cellular location. In the case of ATP, which has a pK_a value of 6.5, the situation is even more complicated, since it will be largely complexed with Mg^{2+} in most tissues, whereas ADP (with one of its pK_a values being 6.3) binds Mg^{2+} less strongly [46] and any 'charge-balanced' reaction scheme that ignores metal complexation would be misleading.

General equations are used in the case of enzymes with broad specificities; the reaction for D-amino acid oxidase (EC 1.4.3.3), for example, being presented as:



Occasionally, Markush structures have been used for reactions, for example, the reaction catalysed by

carbonyl reductase (NADPH; EC 1.1.1.184) is shown as:

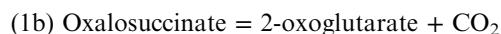
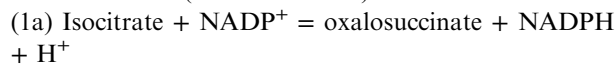
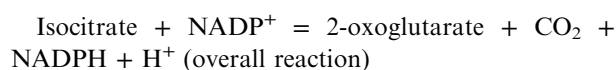


although these have been generally avoided since they are less easy to search.

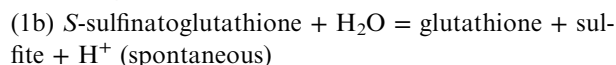
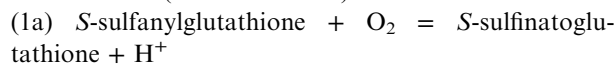
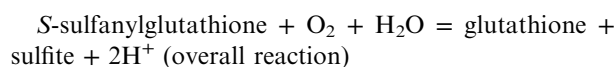
When an enzyme acts on more than one substrate, these may be presented as separate reactions, with those catalysed by aromatic 2-oxoacid reductase (EC 1.1.1.110), for example, being given as:

- (1) (R)-3-(phenyl)lactate + NAD^+ = 3-phenylpyruvate + $\text{NADH} + \text{H}^+$
- (2) (R)-3-(4-hydroxyphenyl)lactate + NAD^+ = 3-(4-hydroxyphenyl)pyruvate + $\text{NADH} + \text{H}^+$
- (3) (R)-(indol-3-yl)lactate + NAD^+ = (indol-3-yl)pyruvate + $\text{NADH} + \text{H}^+$

This is distinguished from reactions that proceed in identifiable steps where the overall reaction is presented and followed by the individual reactions, for example, isocitrate dehydrogenase (NADP^+ ; EC 1.1.1.42) is shown as catalysing:

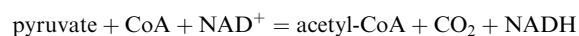


In cases where a product is unstable, and is converted to another in an uncatalysed reaction, this is indicated as being spontaneous, despite objections from purists that all reactions are spontaneous, as in the case of persulfide dioxygenase (EC 1.13.11.18):



Systems and complexes

Some enzymes operate as components of systems or complexes that catalyse a sequence of reactions. The format of ExplorEnz allows it to list complexes of enzymes as a single entry. For example, the pyruvate dehydrogenase system (EC 1.2.1.104) is listed as catalysing:



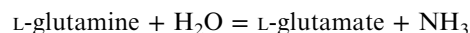
This involves three enzymes: [EC 1.2.4.1](#), pyruvate dehydrogenase (acetyl-transferring), [EC 2.3.1.12](#), dihydrolipoyllysine-residue acetyltransferase and [EC 1.8.1.4](#), dihydrolipoyl dehydrogenase (E3). Each is also classified separately because complexes do not exist in all species and some of the component reactions are also involved in other systems. A single name for such complexes or systems accords with metabolic reality and the way many think about them. Because some systems may involve fusion proteins in some species, the term *System*, rather than complex, is generally preferred. Where one enzyme catalyses two, or more, distinct reactions, these may be listed separately, and it is often found that separate enzymes are involved in some species. It is now recognized that some enzymes may have one or more functions that are distinct from their catalytic activities [47,48]. Such functional promiscuity, which also occurs with noncatalytic proteins [49], is outside the scope of ExplorEnz, which is only concerned with enzyme activities. Indeed, attempting to devise a name that encompassed the multiple functions of some enzymes would have unwieldy results.

Names, names and more names

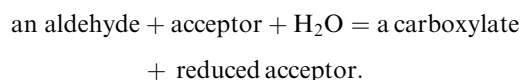
As discussed above, a variety of different names have been sometimes used for the same enzyme and the names by which some enzymes were known gave no clue as to what they catalysed. Changing a name that had already been widely adopted can cause unrest, as was proven when the DPN (diphospho-pyridine nucleotide) was changed to NAD (nicotinamide-adenine dinucleotide) [50]. There are some, who claim to be uninterested in nomenclature, who tend to get very annoyed if the name of their favourite enzyme is changed. It was therefore decided that the name in most common use should be used, provided that it was neither ambiguous nor misleading. This enzyme name was originally termed the *Trivial name* and this was later changed to the *Recommended name*. The latter term provoked some strong reactions from users. For example, Don Nicholson, the designer of the educational metabolic pathways charts [51], asked, on several occasions, how we could be so silly as to name an enzyme pyruvate kinase ([EC 2.7.1.40](#)) when its function was to produce ATP from ADP plus phosphoenolpyruvate. This and similar arguments lead to the name Recommended name being changed to *Accepted name*, indicating that it is name in common use for the enzyme that it is not worth arguing about. Thus, for example, the name catalase gives no indication that the enzyme breaks down hydrogen peroxide

to oxygen and water, but it was too well-established to change. Generally, these names are only altered if they have been shown to be misleading or incorrect, to avoid any unnecessary disruption to textbooks and similar resources.

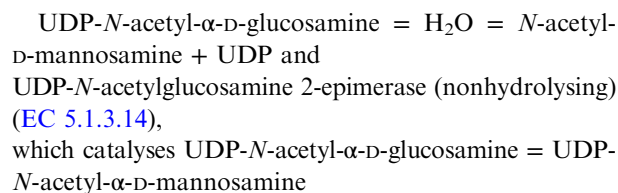
Generally, however, the Accepted name provides an indication of what the enzyme does. For example, -ase after the name of a single substrate such as glutaminase ([EC 3.5.1.2](#)) indicates that it catalyses hydrolysis of the substrate.



Kinase after a substrate name usually indicates that it adds phosphate from NTP to the substrate and dehydrogenase indicated that it oxidizes the substrate by removing H to an acceptor, whereas oxidase indicates that oxygen is the acceptor. Sometimes this simple approach proves to be inadequate and it proves necessary to specify the acceptor in the name. For example, there are several broad specificity aldehyde dehydrogenases that catalyse the general reaction.



In addition to the enzyme that reduces NAD^+ [aldehyde dehydrogenase (NAD^+); [EC 1.2.1.3](#)], there is one that uses NADP^+ [aldehyde dehydrogenase (NADP^+); [EC 1.2.1.4](#)], one that can use either of these (aldehyde dehydrogenase [NAD(P)^+]; [EC 1.2.1.5](#)) and one that uses a quinone as acceptor [aldehyde dehydrogenase (quinone); [EC 1.2.5.2](#)]. The addition of such descriptors in parentheses after the Accepted name has proved necessary to remove ambiguity, as well as to specify what the enzyme actually catalyses. Examples include '(decarboxylating)', '(deaminating)' and '(cyclizing)', which are necessary to clarify the reaction catalysed and to distinguish between similar enzymes. For example, there are two known UDP-*N*-acetylglucosamine 2-epimerases: UDP-*N*-acetylglucosamine 2-epimerase (hydrolysing) ([EC 3.2.1.183](#)), which catalyses



The term (x dependent) is also used to indicate a required reactant. For example, there are two phosphoglucomutases catalysing the overall reaction

α -D-glucose 1-phosphate = D-glucose
6-phosphate

phosphoglucomutase (α -D-glucose-1,6-bisphosphate dependent; EC 5.4.2.2), which requires α -D-glucose 1,6-bisphosphate for activity, and phosphoglucomutase (glucose cofactor) (EC 5.4.2.5), which is activated by glucose.

The qualifying additions to Accepted names can get quite cumbersome, for example, the name D-xylose 1-dehydrogenase (NADP⁺, D-xylono-1,4-lactone forming; EC 1.1.1.424) distinguishes the enzyme catalysing

D-xylose + NADP⁺ = D-xylono-1, 4-lactone + NADPH + H⁺

from D-xylose 1-dehydrogenase (NADP⁺, D-xylono-1, 5-lactone forming) (EC 1.1.1.179), which catalyses

D-xylose + NADP⁺ = D-xylono-1, 5-lactone + NADPH + H⁺.

As far as possible, the additional terms are intended to be self-explanatory and as short as possible since people would be tempted to ignore overcomplicated enzyme names.

Whereas the reactions catalysed are presented in a fixed order in each sub-subclass, the accepted name can be used to give an indication of the direction in which the reaction is perceived to operate. For example, the enzyme that catalyses

L-gulonate + NADP⁺ = D-glucuronate
+ NADPH + H⁺

is named glucuronate reductase (EC 1.1.1.19) to make it clear that the reaction normally proceeds in the direction D-glucuronate reduction.

Other names

As discussed above, many enzymes have been known by alternative names and these are listed for each enzyme entry in the list. The list is intended to be comprehensive so that users can find the enzyme they want by whichever name they feel comfortable with using. For example, a relatively large amount of work had been published on fructose diphosphatase before IUPAC pointed out that a diphosphate was a compound with two phosphates linked together, such as ADP, whereas the substrate for that enzyme was fructose 1,6-bisphosphate. Clearly, it is essential that fructose diphosphatase be included as another name for what is now called fructose-bisphosphatase (EC 3.1.3.11), so that anyone

unfamiliar with the name change can still find the new name by searching.

Where the same name has been used for more than one enzyme, as is the case for xylitol dehydrogenase, which has been used in the literature for D-xylulose reductase (EC 1.1.1.9) and L-xylulose reductase (EC 1.1.1.10), it is marked as ambiguous in other names. Other names that are incorrect or misleading are not excluded but are clearly marked as such. For example, the name formaldehyde dehydrogenase has been quite widely used for S-(hydroxymethyl)glutathione dehydrogenase (EC 1.1.1.284) but the substrate for this enzyme is not formaldehyde, but its adduct with glutathione, S-(hydroxymethyl)glutathione.

Systematic names

Each enzyme is also given a systematic name, which had sufficient chemical information for the structure of the substrate as well as the nature of the reaction to be deduced. Systematic names consist of two parts: the first part contains the name of the substrate or, in the case of a bimolecular reaction catalysed by oxidoreductases, transferases and ligases, of the two substrates separated by a colon; the second part, ending in *-ase*, indicates the nature of the reaction. A number of generic words indicating a type of reaction may be used in either accepted or systematic names: *oxidoreductase*, *oxygenase*, *transferase* (with a prefix indicating the nature of the group transferred), *hydrolase*, *lyase*, *racemase*, *epimerase*, *isomerase*, *mutase* and *ligase*. Where additional information is needed to make the reaction clear, a phrase indicating the reaction or a product should be added in parentheses after the second part of the name, for example, (*ADP-forming*), (*dimerizing*), or (*CoA-acylating*). The systematic names for the lyases take the form 'substrate group-lyase' and 'dehydratase' is used for those enzymes that eliminate water. Detailed rules for devising systematic names, which the faint hearted may prefer to leave to the enzyme nomenclature committee, are set out in the Classification Rules [52], although it should be noted that these are currently being revised.

Systematic names can be quite long and complicated. An example is the systematic name for the enzyme riboflavin synthase (EC 2.5.1.9), 6,7-dimethyl-8-(1-D-ribityl)lumazine:6,7-dimethyl-8-(1-D-ribityl)lumazine 2,3-butanediyltransferase, which is not the easiest name to remember! The Nomenclature Committee of the International Committee of Biochemistry and Molecular Biology (NC-IUBMB) now works closely with the International Union of Pure and Applied Chemistry (IUPAC) as a joint committee on

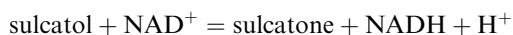
biochemical nomenclature (JCBN) in ensuring that the systematic names accord, as far as practicable, with the IUPAC system [53]. This association of acronyms accommodates two rather different approaches, which do not always sit easily together. The IUPAC name should be unambiguous since it should be possible to convert the name directly into the chemical structure, although the same compound may have more than one, acceptable, IUPAC name depending on the root structure from which it is derived. For example, L-dopa could be 3,4-dihydroxyphenylalanine or (2*S*)-2-amino-3-(3,4-dihydroxyphenyl)propanoic acid, whereas dopamine is 4-(2-aminoethyl)benzene-1,2-diol. Generally, when there is more than one valid alternative name, the IUBMB may indicate their preference.

There have been many arguments to the effect that Systematic names could easily be removed as they are seldom used. Biochemists and pharmacologists often believe that their nonsystematic names are sufficiently well-known not to require definition and IUPAC names are only there as an attempt to make their life miserable. It is much easier to say 'catalase' than to use its systematic name (hydrogen-peroxide:hydrogen-peroxide oxidoreductase) and, as we have seen, systematic names can be much more horrendous than that. However, since the systematic name is intended to define what the enzyme actually catalyses, in terms of substrates used and products formed, it is, like the EC number, an unambiguous identifier of the specific enzyme.

Some of the points discussed above are shown by the results of searching ExplorEnz for a substrate (Fig. 3A), which will return all the enzymes listed as using it, including those enzymes that have been removed or transferred, as shown in Fig. 3B. Although this can be of use in pathway tracing, broad-specificity enzymes that act on this substrate may not be shown unless it is specifically mentioned in the Comments field.

The glossary

The Glossary is used to relate the common names of the compounds shown in the reaction field or accepted name with their IUPAC names and any alternative names that may be used. Thus, for example, the reaction catalysed by sulcatone reductase (EC 1.1.1.260) is shown as.



and the accompanying glossary entry is as follows:
sulcatone = 6-methylhept-5-en-2-one; sulcatol=6-methylhept-5-en-2-ol.

The Glossary can be accessed separately at <http://www.enzyme-database.org/glossary.php>, where the entries are each linked to the ChemSpider database [54] to allow access to their structures and other chemical information. A list of the symbols and abbreviations used is also available at <https://www.enzyme-database.org/abbrev.php>.

Diagrams

Several of the enzyme entries are linked to diagrams that show the involvement of the enzyme in a metabolic pathway and/or its reaction mechanism. These diagrams were developed by H.B.F. Dixon and G.P. Moss and are now being maintained and expanded by G.P. Moss. A complete list of these is available at <https://iubmb.qmul.ac.uk/enzyme/reaction/>.

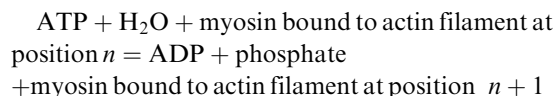
Links to other databases

As the information provided in ExplorEnz is largely restricted to that necessary for classification and nomenclature, these links are provided to facilitate access to more information. The involvement of individual metabolic pathways can, for example, be found in the KEGG [12] and MetaCyc [55] databases, kinetic data are available through BRENDA [13] and the thermodynamics of individual reactions can be found in the NIST database [40], which is titled GTD in the *Links* field.

New subclasses

The macromolecular isomerases

The ATPases that were listed as hydrolases contained some enzymes whose function was to change the conformation of large molecules, such as myosin ATPase, which was classified as EC 3.6.1.32 and subsequently moved to EC 3.6.4.1. This, however, kept these enzymes inappropriately listed amongst enzymes whose primary function was hydrolysis, so it has been decided to group those enzymes that altered macromolecular conformation into a subclass of the isomerases, EC 5.6 *Isomerases altering macromolecular conformation*, with those altering polypeptide conformation or assembly being grouped in EC 5.6.1 and those altering nucleic acid conformation in EC 5.6.2. Myosin ATPase has found its last resting place, we hope, as EC 5.6.1.8, with the reaction being given as:



A**Simple search**

For advanced search option, [click here](#).

Search for in

☐ all fields ☐ Use regular expressions [\[what are these?\]](#) ☐ Full-text search

or

☒ select fields:

☐ EC Number	☐ Comments
☒ Accepted name	☐ References
☒ Reaction	☐ PubMed ID
☒ Other name(s)	☐ Glossary
☐ Systematic name	

and display ☐ (highlighting matches)

☐ all fields

or

☒ select fields:

☒ Accepted name	☐ Comments
☒ Reaction	☐ Links to other databases
☒ Other name(s)	☐ References
☒ Systematic name	☐ History
☐ Glossary	

Sort results by , displaying entries per page.

B**EC 1.4.3.6**

Deleted entry: amine oxidase (copper-containing). This was classified on the basis of cofactor content rather than reaction catalysed and is now known to contain two distinct enzyme activities. It has been replaced by two enzymes, [EC 1.4.3.21](#) (primary-amine oxidase) and [EC 1.4.3.22](#) (diamine oxidase)

EC 1.4.3.21

Accepted name: primary-amine oxidase

Reaction: $\text{RCH}_2\text{NH}_2 + \text{H}_2\text{O} + \text{O}_2 = \text{RCHO} + \text{NH}_3 + \text{H}_2\text{O}_2$

Other name(s): amine oxidase (*ambiguous*); amine oxidase (copper-containing); amine oxidase (pyridoxal containing) (*incorrect*); benzylamine oxidase (*incorrect*); CAO (*ambiguous*); copper amine oxidase (*ambiguous*); Cu-amine oxidase (*ambiguous*); Cu-containing amine oxidase (*ambiguous*); diamine oxidase (*incorrect*); diamino oxyhydrase (*incorrect*); histamine deaminase (*ambiguous*); histamine oxidase (*ambiguous*); monoamine oxidase (*ambiguous*); plasma monoamine oxidase (*ambiguous*); polyamine oxidase (*ambiguous*); semicarbazide-sensitive amine oxidase (*ambiguous*); SSAO (*ambiguous*)

Systematic name: primary-amine:oxygen oxidoreductase (deaminating)

EC 1.4.3.22

Accepted name: diamine oxidase

Reaction: $\text{histamine} + \text{H}_2\text{O} + \text{O}_2 = (\text{imidazol-4-yl})\text{acetaldehyde} + \text{NH}_3 + \text{H}_2\text{O}_2$

Other name(s): amine oxidase (*ambiguous*); amine oxidase (copper-containing) (*ambiguous*); CAO (*ambiguous*); Cu-containing amine oxidase (*ambiguous*); copper amine oxidase (*ambiguous*); diamine oxidase (*ambiguous*); diamino oxyhydrase (*ambiguous*); histaminase; histamine deaminase (*incorrect*); semicarbazide-sensitive amine oxidase (*incorrect*); SSAO (*incorrect*)

Systematic name: histamine:oxygen oxidoreductase (deaminating)

EC 2.1.1.8

Accepted name: histamine *N*-methyltransferase

Reaction: $\text{S-adenosyl-L-methionine} + \text{histamine} = \text{S-adenosyl-L-homocysteine} + \text{N-methylhistamine}$

Other name(s): histamine 1-methyltransferase; histamine methyltransferase; histamine-methylating enzyme; imidazolemethyltransferase; S-adenosylmethionine-histamine *N*-methyltransferase

Systematic name: S-adenosyl-L-methionine:histamine *N*-methyltransferase

EC 4.1.1.22

Accepted name: histidine decarboxylase

Reaction: $\text{L-histidine} = \text{histamine} + \text{CO}_2$

Other name(s): L-histidine decarboxylase; L-histidine carboxy-lyase

Systematic name: L-histidine carboxy-lyase (histamine-forming)

Fig. 3. Searching the Enzyme List. A. A simple search within ExplorEnz for 'histidine' showing the Fields selected for the search and display. B. The output of the simple search in (A). The last enzyme found, γ -glutamylhistamine synthase ([EC 6.3.2.18](#)), has been omitted for brevity.

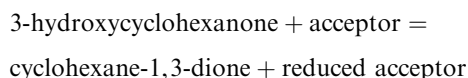
Table 4. Numbers of enzymes in each EC class of the NC-IUBMB list (as of October 2021).

Class	1	2	3	4	5	6	7	All
Current	1936	1967	1322	716	306	239	95	6581
Transferred	421	87	364	79	13	6	0	970
Deleted	93	84	112	31	10	9	0	339
Total	2450	2138	1798	826	329	254	95	7890

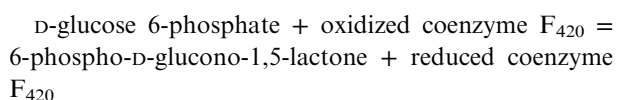
The history field in the entry for this enzyme reads [EC 5.6.1.8 created 1984 as EC 3.6.1.32, transferred 2000 to EC 3.6.4.1, transferred 2018 to EC 5.6.1.8].

The problematical 99s

The occurrence of 99 in the subclass or sub-subclass of an EC number implies that it is not clear where else to put the enzyme. In some cases, this was from a reluctance to clutter-up the enzyme list with subclasses or sub-subclasses that contained only one or two enzymes. Some of the others were historical because of limitations of enzyme assay procedures. Amongst the oxidoreductases, there were several enzymes where the natural acceptor was unknown. These are classified as EC 1.-.99.-, *With unknown physiological acceptors*, or simply *Miscellaneous*, in EC 1.13 and EC 1.14. Examples include 3-hydroxycyclohexanone dehydrogenase (EC 1.1.99.26) and the reaction is given in the general form:



where the Comments state, '2,6-Dichloroindophenol and methylene blue can act as acceptors'. Although it might be argued that these enzymes should not have been classified until the physiological acceptor had been established, they have been there from the start of the Enzyme list, with the intention of moving them when the acceptor was identified. This subclass, however, also included enzymes where the acceptor was known but was not common enough to justify developing a new sub-subclass for it. These included glucose-6-phosphate dehydrogenase (coenzyme-F₄₂₀), which catalyses



This had been classified as EC 1.99.34, but it was messy to include enzymes with known acceptors together with those where the acceptor had not been

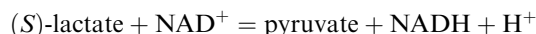
identified in the same group. This has now been resolved by the creation of new sub-subclass(es) EC 1.-.98.-, 'With other, known, physiological acceptors', to which glucose-6-phosphate dehydrogenase (coenzyme-F₄₂₀) has been moved as EC 1.1.98.2, with oxidized coenzyme F₄₂₀ being listed in the Glossary as *N-(N-{O-[5-(8-hydroxy-2,4-dioxo-2,3,4,10-tetrahydropyrimido[4,5-*b*]quinolin-10-yl)-5-deoxy-L-ribityl-1-phospho]-(S)-lactyl}-γ-L-glutamyl)-L-glutamate*. It has, so far, been possible to assign 78 enzymes to this group. In accordance with this change, the EC 1.x.99.y sub-subclass has been renamed 'With unknown, physiological acceptors', with the hope that future work will result in its content being further reduced. The effects of such 'cleaning-up' operations are summarized in Table 4.

The removal of some other '99 sub-subclass' hydrolases to the translocase and isomerases has already been discussed and other movements can be considered, to further increase the clarity of the system. For example, 2.4.99.x in the glycosyltransferases subclass (EC 2.4) contains 20 enzymes, but there may be sufficient sialyltransferases amongst these to justify the generation of a separate sub-subclass.

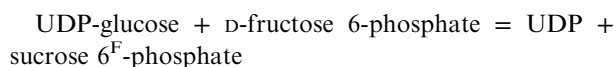
Clarifying chirality

When the Enzyme List was first compiled, it was done so by biochemists with biochemists as the intended users and at that time they had some quaint ideas about chirality, which those in other disciplines might find confusing. In fact, the 1992 version of the classification rules states that 'The prefix D- should be omitted for all D-sugars and L- for individual amino acids, unless ambiguity would be caused'. Thus, if an amino acid were named, it could be assumed that it was an L-amino acid, except of course in the case of glycine. Similarly, if it were a sugar it would be a D-sugar unless otherwise stated. Thus, Accepted names include lysine dehydrogenase (1.4.1.15) and lysine decarboxylase (EC 4.1.1.18), although the *Reactions* fields in these cases do specify that it is L-lysine that is involved. This strange convention has become

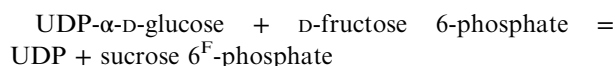
extended to derivatives of amino acids and sugars, although there is some lack of clarity as to when a compound ceases to be a derivative. This ambiguity is shown by the entry for L-lactate dehydrogenase (EC 1.1.1.27) where the reaction is shown as:



In the case of glucose and its derivatives, the anomeric configuration has now also been indicated, where appropriate. For example, the reaction catalysed by sucrose-phosphate synthase (EC 2.4.1.14) has been changed from



to



Conclusions

It used to be claimed that it took a full year to paint the Forth Bridge in Scotland and, when the end was reached, it was time to start at the beginning again. Whether maintaining and expanding the enzyme list may be compared with that, it is certainly work that is unlikely ever to reach a conclusion. The list will continue to expand as new enzymes that meet the inclusion criteria are reported, and further information on existing entries requires its revision. The addition of new EC classes remains a possibility and new subclasses (and sub-subclasses) should certainly be necessary in the future.

The continued value of the Enzyme lists depends on input from the community. Proposals for enzymes to be included may be made online through <https://www.enzyme-database.org/newform.php>. Errors in existing entries or suggestions for updates to these entries may be through <https://www.enzyme-database.org/updateform.php>. We also welcome suggestions on the nature of the information provided and the applicability of the EC system.

Conflict of interest

The authors declare no conflict of interest.

Author contributions

Both authors contributed equally to the preparation of this manuscript.

Data accessibility statement

The IUBMB Enzyme List can be accessible at the ExplorEnz website, <https://www.enzyme-database.org>.

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